

CANDIDATE'S NAME:

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REGISTRATION NUMBER:

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553/1

BIOLOGY (Theory)

Paper 1

JULY/AUGUST 2026

2 Hours



KAMPALA EXCELLENCE EXAMINATION BOARD (KEEB)

JOINT MOCK 2026

Uganda Certificate of Education

BIOLOGY (THEORY)

Paper 1

2 Hours

INSTRUCTIONS TO CANDIDATES

- This examination paper consist of **two** sections **A** and **B**
- Section **A** has **three** compulsory items.
- Responses to section **A** must be written in the spaces provided.
- Section **B** has two parts **I** and **II**, respond to only one item in each part.
- Responses to section **B** must be written in the answer booklets provided.
- Respond to **five** items in all.
- Any additional items responded to **will not** be scored

SECTION A

Respond to **all** the items in this section in the space provided

ITEM 1

Mr. Lwanga is a vegetable farmer in a village near a busy stone-quarrying site in Nakasongola District. Recently, he has noticed that his cabbages are growing very slowly and have turned a yellowish-green colour. He also observed that even after a light rain, the plants appear to wilt by midday.

A close inspection reveals that the surfaces of Mr. Lwanga's cabbage leaves are covered in a thick layer of white stone dust from the quarry. Additionally, the soil in his garden has become very compacted due to heavy machinery moving nearby, which he suspects is affecting how the roots behave. The quarrying was stopped for one month for a holiday, fortunately it rained heavily on Lwanga's garden, this made him to weed his garden, cabbage leaves started turning green and he managed to realize some yield

Task

(a) Explain the effect of quarrying on Lwanga's cabbage

(b) Explain how the cabbage managed to yield after quarry holiday

ITEM 2

Okello is a talented Senior Four student who recently joined a local community football team. To keep up with the older players, his friends introduced him to "Khat" (Mairungi) and "Kuber," claiming these substances would give him "limitless energy" and keep him alert during matches without training. However, during a high-stakes regional tournament, Okello's performance took a worrying turn.

Ten minutes into the second half, Okello movements became uncoordinated, complains of thirst, and increased heartbeat. Suddenly, he collapsed on the pitch crying his left calf muscle had become hard, tight, and extremely painful. He had to be carried off the field, unable to continue. The team medic noticed Okello was severely dehydrated and trembling.

Task

(a) Explain the cause of Okello's experience during the game

(b) Advise Okello on how he can manage the condition that he is experiencing

ITEM 3

In the rural village of Ajonyi, a recent community health survey revealed that many families are struggling with reproductive health challenges. Robert and his wife, Grace, are expecting their fourth child. Robert has been expressing concern, hoping for a son to "**continue the family name**," and he has been blaming Grace for their previous three daughters.

They were advised to visit a professional counselor who then told them children's sex are determined by the biological father, Robert became annoyed that Grace bribed the counselor to talk against him. Due to terrible words that Robert told his wife Grace, she became unfaithful and started having affairs outside marriage, became pregnant and abandoned going for ante natal care, contracted syphilis and HIV/AIDS. As a result, she gave birth to a baby boy unfortunately the boy was also tested positive for HIV and her husband as well. Meanwhile, the village health team has noted an increase in reproductive health complications among teenagers, including undiagnosed infections and early pregnancy risks.

Task

Using your knowledge of genetics,

- (a) explain to Robert how sex of children is determined by the biological father

- (b) Explain;

- (i) the likely effect of Grace's life style to her family

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- (ii) how she can overcome the challenges she is facing and that of her child
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SECTION B

PART 1

*Respond to only **one** item from this part*

Response should be written in the answer booklet(s) provided

ITEM 4

In Aboket village, the local wetland has traditionally provided clean water, mudfish, and a breeding ground for the Grey Crowned Crane. However, over the last 15 years, the village population has grown rapidly. To sustain their families, many residents have turned to large-scale sand mining and the conversion of the wetland into rice paddies.

The local environment officer has noticed a sharp decline in biodiversity. The water has become dark cloudy and shallow, and the local fishermen are struggling to find enough fish to sell. A census of the Grey Crowned Crane shows that nesting pairs have dropped from 40 to only 4 in the last decade. The community is now facing a shortage of clean water and a loss of their primary source of protein.

The table 1 below shows data collected by the environment officer over the years.

Table 1

Year	Human Population (thousands)	Monthly Fish Yield (kg)
2010	1.2	850
2015	2.8	580
2020	4.5	310
2024	6.7	95

Population and Fish Yield in Aboket Village

Task

- (a) Using the data provided in the table explain;

- (i) the relationship between the human population growth and the fish yield in Aboket village
- (ii) the cause of the challenges experienced by the community
- (b) Explain the measures the residents should take to reduce the challenges reported by the environment officer

ITEM 5

In the fast-growing town of Namanve, a large open-air market has become a significant source of environmental concern. Every week, tons of waste are discarded in an unprotected area behind the market stalls. This waste pile consists of a variety of materials that include; plastics old phones, batteries, old shoes, old clothes and rotting food materials.

Mr. Mukasa, a local farmer whose garden is located directly downhill from this waste site, has observed a sharp decline in his crop productivity. His maize plants appear stunted with yellowing leaves (chlorosis), and the soil has become increasingly difficult to plough because of thick layers of buried plastic. During the rainy season, dark-coloured liquid (leachate) from the waste pile flows directly into his garden.

Task

- (a) Explain how the different categories of waste at the market site are contributing to the degradation of the soil structure and fertility in Mr. Mukasa's garden
- (b) Suggest sustainable waste management strategy for the Namanve market that would help the community achieve the Sustainable Development Goal of "Life on Land."

Part II

*Respond to only **one** item from this part*

Response should be written in the answer booklet(s) provided

ITEM 6

St. Mark Secondary School recently organized a "**Health and Wellness Fair**" to sensitize the surrounding community about common lifestyle diseases. As part of the Biology project, the S4 students were tasked to demonstrate how the body processes nutrients and why health screenings specifically urine analysis are critical for early detection of metabolic imbalances.

During demonstration, a 45-year-old local shopkeeper, Mr. Opio, who complained of persistent fatigue, frequent thirst after the meal of sweet potatoes and eggs, also complains of rapid weight loss. Provided his medical report papers to students to explain to him. On the report, the following lab results were written

Lab report

Test	Results/observation
Benedict test on urine (presence of glucose)	Turned brick-red
Biuret test (presence of proteins)	Turned purple

The students then had to explain to the local residents not only how these substances reached the urine but also the deeper physiological implications of their presence in a person's excretory system.

Task

- (a) Explain to the residents how nutrients in Mr. Opio's meal reached urine
- (b) Explained to Mr. Opio the cause of what he is experiencing and advise him on how to manage the conditions

ITEM 7

Mr. Balu is a 50-year-old taxi driver operating in the busy, dusty streets of Kampala. To cope with the long, stressful hours, he smokes two packets of cigarettes daily. His diet consists mostly of "Rolex" (fried chapatti and eggs) and deep-fried chicken from roadside stalls, which he washes down with sugary sodas.

For the past six months, Mr. Balu has developed a persistent, "wet" cough that refuses to go away. Despite his high-calorie diet, he is noticeably losing weight and feels constantly weak. Recently, he started experiencing sharp chest pains and breathlessness whenever he climbs the stairs to his apartment.

Task

- (a) Explain the cause of Balu's health conditions
- (b) Suggest the possible ways Balu can manage his health status

END